

Task Manager (Desktop GUI)

1. Setup Instructions

1. Create virtual environment:

```
python -m venv .venv
```

2. Activate:

- Windows:

```
.venv\Scripts\activate
```

- macOS/Linux:

```
source .venv/bin/activate
```

3. Install dependencies:

```
pip install matplotlib
```

4. Run the application:

```
python main.py
```

5. Freeze requirements:

```
pip freeze > requirements.txt
```

2. Features

- Add, Delete tasks
 - Toggle task status (Pending ↔ Completed)
 - Task prioritization (High, Medium, Low)
 - GUI-based task management using Tkinter
 - Task completion report with pie chart visualization
 - Persistent storage using JSON file
-

3. OOP Design

Classes Used:

1. **Task (Entity)**
 - Stores task details (title, priority, status)
 2. **FileHandler (Storage)**
 - Handles saving/loading tasks to JSON
 3. **TaskManager (Service/Manager)**
 - Manages task operations (add, delete, toggle, stats)
 4. **TodoApp (UI Controller)**
 - Handles user interface and interaction
-

Concepts:

- Encapsulation
 - Modular Design
 - Separation of Concerns
 - Object-Oriented Programming (OOP)
-

4. Functionality

The system allows users to manage daily tasks efficiently using a graphical interface.

Users can:

- Add new tasks with priority
- View all tasks in a structured table
- Mark tasks as completed
- Delete tasks
- View completion statistics

All operations are smoothly handled through the GUI and backend logic.

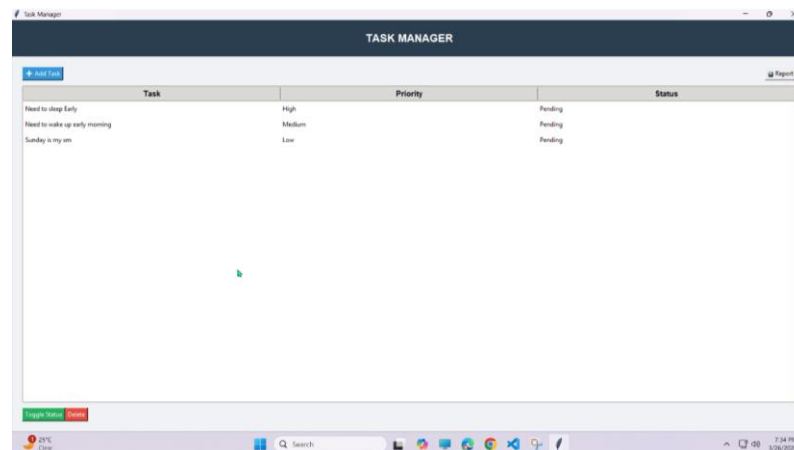
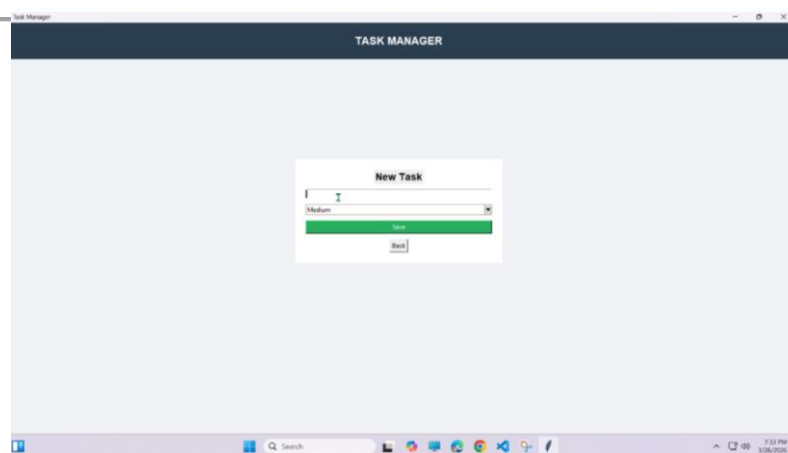
5. Persistence

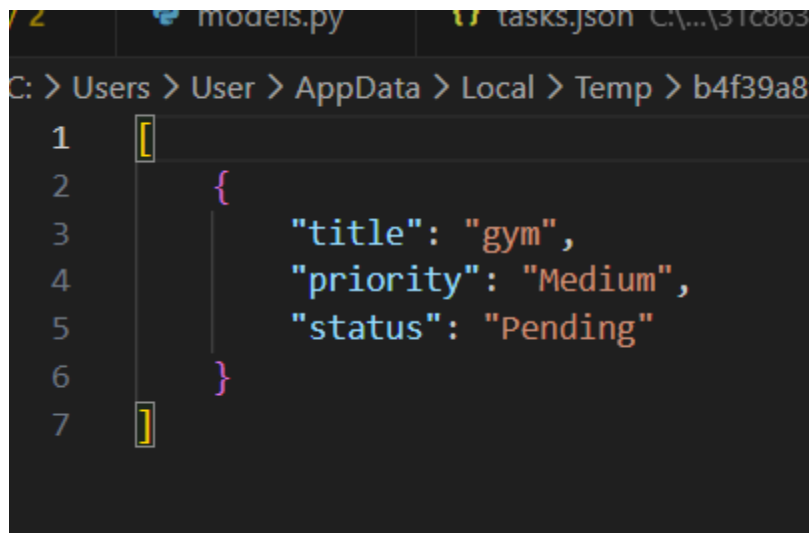
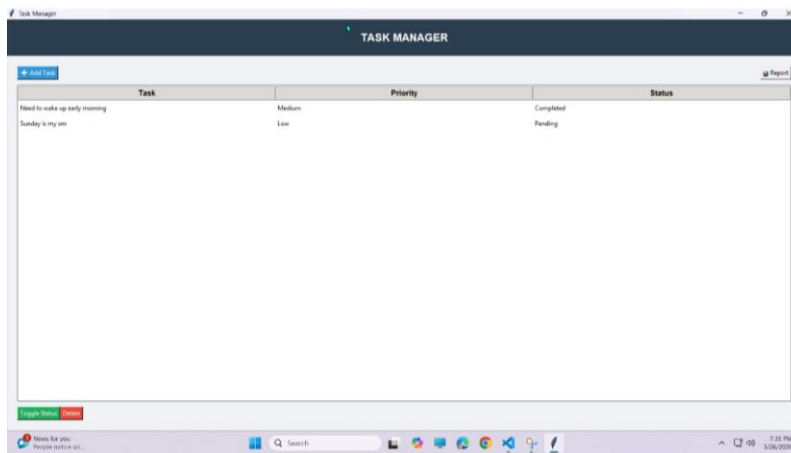
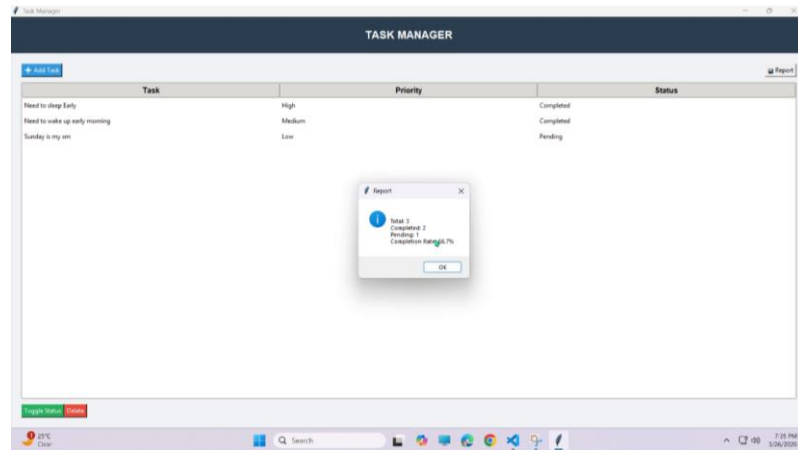
- Data is stored in **tasks.json** file
 - Automatically loads tasks on startup
 - Saves data after every operation
 - Uses JSON format for simplicity and readability
-

6. Validation & Error Handling

- Prevents empty task input
- Handles missing selections safely
- Confirms before deleting tasks
- Displays user-friendly warning messages
- Avoids application crashes

7. Screenshots (Add Manually)





8. Conclusion

This project successfully demonstrates a complete desktop-based Task Manager application.

It shows strong understanding of:

- Python programming
- GUI development using Tkinter
- Data visualization using Matplotlib
- File handling with JSON
- Object-Oriented Design principles

The application is simple, user-friendly, and efficient for managing daily tasks.